Unit: Unit 9: Data Statistical Analysis **Name:** ________________
Topic: Measures of Central Tendency **Date:** ____________

Advanced (Habanero)

Q1. What is the mean of the numbers $2, 4, 6$?
(A) 2
(B) 4
(C) 6
(D) 3
(E) 12

Q2. Which of the following is a measure of central tendency?
(A) Range
(B) Mode
(C) Variability
(D) Standard Deviation
(E) Interquartile Range

Q3. What is the median of the numbers $1, 3, 5, 7, 9$?
(A) 1
(B) 3
(C) 5
(D) 7
(E) 9

Q4. A dataset has a mean of 10 and a median of 12. What can be said about the dataset?
(A) It is symmetric
(B) It is skewed to the right
(C) It is skewed to the left
(D) It has no mode
(E) It has no range

Q5. Which of the following datasets has the largest mode?
(A) $1, 1, 2, 3$
(B) $2, 2, 2, 4$
(C) $3, 3, 3, 5$
(D) $4, 4, 4, 6$
(E) $5, 5, 5, 7$

Q6. What is the mode of the numbers $2, 4, 4, 6, 6, 6$?
(A) 2
(B) 4
(C) 6
(D) 3
(E) 5

Q7. A dataset has a mean of 8 and a standard deviation of 2. What is the z-score of the value 10?
(A) -1
(B) 0
(C) 1
(D) 2
(E) 3

Q8. Which of the following is NOT a characteristic of the mean?
(A) It is sensitive to outliers
(B) It is a measure of central tendency
(C) It is always an integer
(D) It can be used for skewed datasets
(E) It can be used for datasets with missing values

Open Ended Questions (FRQ)

Q9. Calculate the mean, median, and mode of the dataset $2, 4, 6, 8, 10$. Compare the measures of central tendency and discuss the implications of each.

Q10. Suppose we have two datasets: $1, 3, 5, 7, 9$ and $2, 4, 6, 8, 10$. Calculate the mean, median, and mode of each dataset. Compare the measures of central tendency between the two datasets and discuss any similarities or differences.

Chef's Tips

- Read each question carefully
- Use the space provided for calculations
- Check your work before submitting